

LIMESTONE AND KARST LANDFORMS / INDIA CAVES-MEGHALAYAN AGE

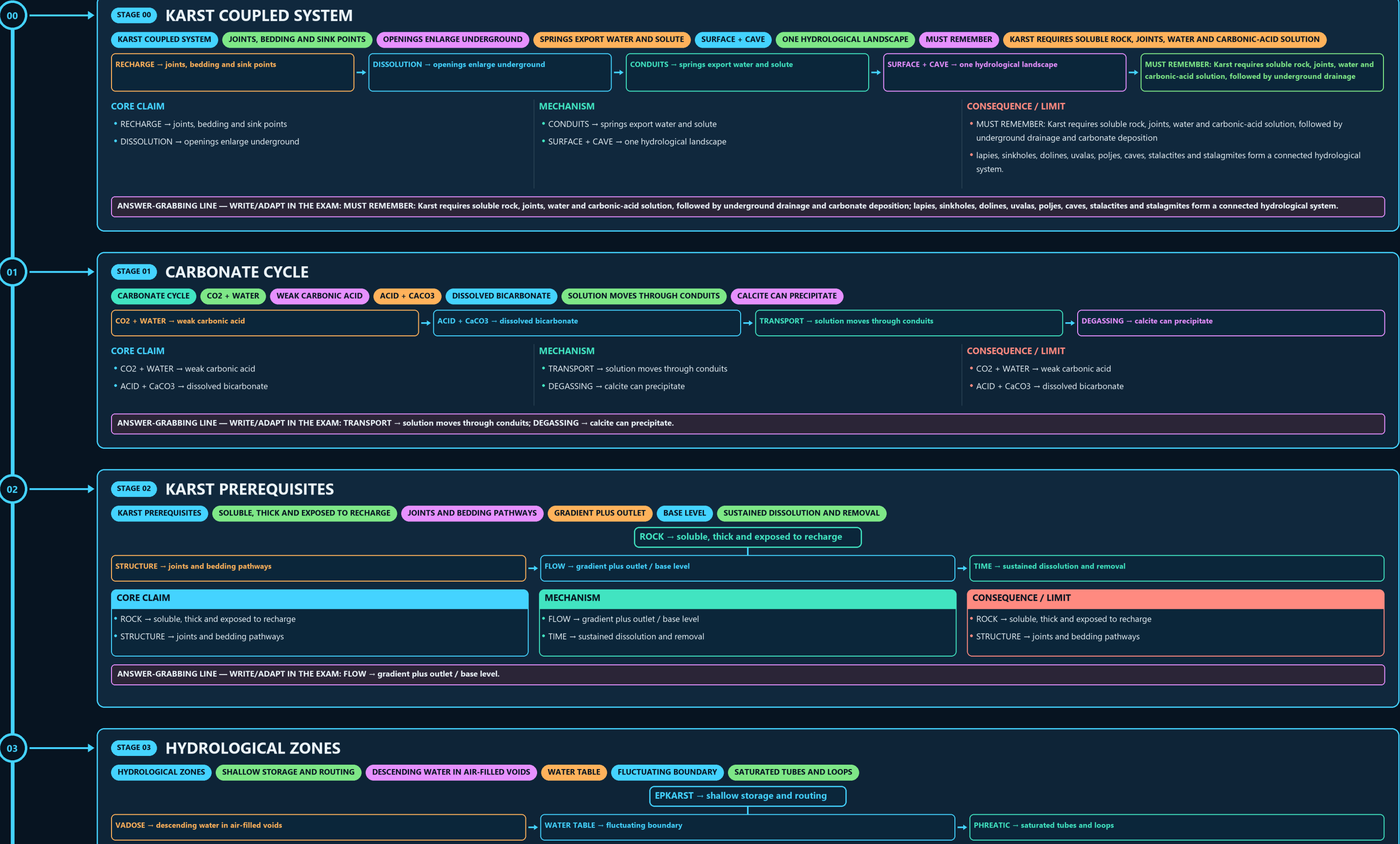
KARST COUPLED SYSTEM → CARBONATE CYCLE → KARST PREREQUISITES → HYDROLOGICAL ZONES → SURFACE-KARST LADDER → CAVE-DEVELOPMENT SEQUENCE

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

PRIMARY CORE FLOW SUBORDINATE EXTRA APPROVAL / REVIEW

Approval: FALSE • Pending user review • source generation g5 and all prior artifacts unchanged

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles



ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: LATE CHANGE → collapse, fill or renewed solution.

06

STAGE 06 SPELEOTHEM MAP

SPELEOTHEM MAP ROOF DRIP DEPOSIT FLOOR SPLASH DRIP DEPOSIT ROOF AND FLOOR FORMS JOIN FILM FLOW ON WALL OR SLOPE

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
STALACTITE → roof drip deposit	COLUMN → roof and floor forms join	STALACTITE → roof drip deposit
STALAGMITE → floor splash / drip deposit	FLOWSTONE / CURTAIN → film flow on wall or slope	STALAGMITE → floor splash / drip deposit
CORE CLAIM - STALACTITE → roof drip deposit - STALAGMITE → floor splash / drip deposit	MECHANISM - COLUMN → roof and floor forms join - FLOWSTONE / CURTAIN → film flow on wall or slope	CONSEQUENCE / LIMIT - STALACTITE → roof drip deposit - STALAGMITE → floor splash / drip deposit

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: FLOWSTONE / CURTAIN → film flow on wall or slope.

07

STAGE 07 INDIA CAVE FIREWALL

INDIA CAVE FIREWALL MEGHALAYA LIMESTONE NATURAL CAVES IN ANDHRA PRADESH HUMAN ROCK-CUT ARCHITECTURE LOCATION DOES NOT REPLACE GENETIC CLASSIFICATION

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
MAWMLUH / MAWSMAI / SIJU → Meghalaya limestone	AJANTA / ELLORA → human rock-cut architecture	MAWMLUH / MAWSMAI / SIJU → Meghalaya limestone
BORRA / BELUM → natural caves in Andhra Pradesh	RULE → location does not replace genetic classification	BORRA / BELUM → natural caves in Andhra Pradesh
CORE CLAIM - MAWMLUH / MAWSMAI / SIJU → Meghalaya limestone - BORRA / BELUM → natural caves in Andhra Pradesh	MECHANISM - AJANTA / ELLORA → human rock-cut architecture - RULE → location does not replace genetic classification	CONSEQUENCE / LIMIT - MAWMLUH / MAWSMAI / SIJU → Meghalaya limestone - BORRA / BELUM → natural caves in Andhra Pradesh

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → location does not replace genetic classification.

08

STAGE 08 PROXY PIPELINE

PROXY PIPELINE RAINFALL, VEGETATION AND SOIL CO2 ROUTE AND RESIDENCE TIME ELEMENT SIGNAL DATED SERIES WITH UNCERTAINTY

CLIMATE → rainfall, vegetation and soil CO2	→	RECHARGE → route and residence time	→	CALCITE → isotope / element signal	→	CHRONOLOGY → dated series with uncertainty
CORE CLAIM - CLIMATE → rainfall, vegetation and soil CO2 - RECHARGE → route and residence time		MECHANISM - CALCITE → isotope / element signal - CHRONOLOGY → dated series with uncertainty		CONSEQUENCE / LIMIT - CLIMATE → rainfall, vegetation and soil CO2 - RECHARGE → route and residence time		

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CALCITE → isotope / element signal; CHRONOLOGY → dated series with uncertainty.

09

STAGE 09 MEGHALAYAN HIERARCHY

MEGHALAYAN HIERARCHY KM-A SPELEOTHEM AT MAWMLUH CLOSE DISTINCTION SOLUTION ENLARGES VOIDS WHEREAS DEPOSITION BUILDS SPELEOTHEMS; STALACTITES HANG MEGHALAYAN CHRONOSTRATIGRAPHIC UNIT ARE CATEGORICALLY DIFFERENT.

HOLOCENE → Epoch	→	MEGHALAYAN → Age / Stage	→	GSSP → KM-A speleothem at Mawmluh
CORE CLAIM - QUATERNARY → Period - HOLOCENE → Epoch - MEGHALAYAN → Age / Stage		MECHANISM - GSSP → KM-A speleothem at Mawmluh - CLOSE DISTINCTION: Solution enlarges voids whereas deposition builds speleothems - stalactites hang, stalagmites rise, columns join them, and Meghalaya caves, Bhimbetka shelters and the		CONSEQUENCE / LIMIT Meghalayan chronostratigraphic unit are categorically different.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CLOSE DISTINCTION: Solution enlarges voids whereas deposition builds speleothems; stalactites hang, stalagmites rise, columns join them, and Meghalaya caves, Bhimbetka shelters and the.

CORE CLAIM

- QUATERNARY → Period
- HOLOCENE → Epoch
- MEGHALAYAN → Age / Stage

MECHANISM

- GSSP → KM-A speleothem at Mawmluh
- CLOSE DISTINCTION: Solution enlarges voids whereas deposition builds speleothems
- stalactites hang, stalagmites rise, columns join them, and Meghalaya caves, Bhimbetka shelters and the

CONSEQUENCE / LIMIT

- Meghalayan chronostratigraphic unit are categorically different.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CLOSE DISTINCTION: Solution enlarges voids whereas deposition builds speleothems; stalactites hang, stalagmites rise, columns join them, and Meghalaya caves, Bhimbetka shelters and the.

CORE CLAIM

- 4.2 ka EVENT → correlation guide
- GSSP POINT → physical lower boundary

MECHANISM

- MEGHALAYAN → named formal interval
- ANTHROPOCENE → not a ratified Epoch

CONSEQUENCE / LIMIT

- EVIDENCE LIMIT: Cave age, host-rock age, occupation age and formal geological-age boundary must not be collapsed
- tourism and infrastructure effects depend on recharge, passage geometry, biodiversity and carrying capacity.

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- 4.2 ka EVENT → correlation guide
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ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: EVIDENCE LIMIT: Cave age, host-rock age, occupation age and formal geological-age boundary must not be collapsed; tourism and infrastructure effects depend on recharge, passage geometry, biodiversity and carrying capacity.

CORE CLAIM

- MAP → recharge, passages, springs and values
- AVOID → irreversible quarry and pollution pathways

MECHANISM

- MANAGE → visitation, waste and water use
- MONITOR → hydrology, biodiversity and structural change

CONSEQUENCE / LIMIT

- MAP → recharge, passages, springs and values
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ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: MANAGE → visitation, waste and water use; MONITOR → hydrology, biodiversity and structural change.

OPTIONAL SOURCE DEPTH

- Meghalaya has India's premier karst systems. Krem Liat Prah is widely cited as India's longest
- The Cherrapunji limestone cave shows well-developed stalactite and stalagmite formations (dripstone deposition).
- Cave tourism spans two different categories: natural karst caves in Meghalaya and rock-cut/
- Very high rainfall over thick limestone makes Meghalaya ideal for rapid solution and cave growth.

ADVANCED DEBATE

- These caves double as palaeoclimate archives — their stalagmites record past rainfall/drought.
- GSSP = Global Boundary Stratotype Section and Point — the physical 'golden spike' defining a time boundary.
- The Meghalayan Age is the youngest stage of the Holocene (last ~4,200 years); its GSSP is the KNI-51 stalagmite in Mawmluh Cave, Meghalaya.
- The stalagmite isotope signal marks abrupt weakening of the Indian summer monsoon around 4.2 ka.

LEGACY / NUANCE

- Krem Liat Prah in Meghalaya is widely cited as India's longest natural cave; mapped length is survey-dependent.
- Cherrapunji limestone cave: developed stalactite/stalagmite (Khullar).
- Ajanta (Maharashtra), Pachmarhi (MP ~500 caves), Khandagiri/Udayagiri (Odisha).
- High rainfall + thick limestone = rapid karst solution in Meghalaya.

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.